

ZITIAntao Lin

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EDUCATION

Northeastern University

Master of Science in Information Systems (GPA: 3.5/4.0)

Boston, MA

Jan 2025 – Expected Dec 2026

Civil Aviation Flight University of China

Bachelor of Engineering in Information and Computing Science

Sichuan, China

Sept 2020 – Jul 2024

TECHNICAL SKILLS

Languages: Python (Expert), C++, Java, SQL, C#, MATLAB, LaTeX

AI & Machine Learning: LLMs, MLLM, Computer Vision, Transformer Models, Time Series Forecasting, Depth Generation

Tools & Frameworks: PyTorch, Unity, ROS, XR/Mixed Reality (MR), Git, Docker, TensorRT, HPC (Slurm/Explorer)

Engineering Practices: Data Pipelines, Modular Frameworks, Funnel Analysis, System Design, Code Review, HPC Optimization

PROFESSIONAL EXPERIENCE

Northeastern University — CRAFT Lab

Graduate Research Assistant (Advisor: Prof. Gilbert Yang Ye)

Boston, MA

Feb 2025 – Present

Design and implement end-to-end interactive robotics systems that integrate complex perception, learning, and control modules.

Develop Large Language Model (LLM) driven components for human-robot interaction and real-time XR/Unity simulation tools.

Integrated heterogeneous data sources including vision features and gaze signals into unified, debuggable processing pipelines.

Build real-time simulation and visualization tools for embodied AI and HRI experiments to validate system-level performance.

Collaborate with interdisciplinary researchers to translate research hypotheses into scalable and maintainable software architectures.

Northeastern University

Teaching Assistant — INFO 6105: Data Science Engineering Methods

Boston, MA

Jan 2026 – Present

Lead technical support for data processing pipelines and machine learning engineering workflows in Python for over 100 students.

Review and debug student code implementation, providing structured feedback on algorithmic efficiency and reproducibility.

Conduct office hours to facilitate problem-solving for complex analytical workflows and foundational machine learning models.

Beijing Sohu New Media Information Technology Co Ltd

Software Engineering & Product Operations Intern (Mobile Applications)

Beijing, China

Mar 2021 – Aug 2021

Optimized features for a large-scale mobile application by collaborating with engineering teams to enhance system functionality.

Conducted deep-dive analysis of user behavior and traffic data using funnel analysis to identify and resolve usage bottlenecks.

Maintained and enhanced components of a modular framework, supporting rapid feature iteration and scalable development.

Translated complex product requirements into actionable technical insights within a high-pressure production environment.

PROJECTS EXPERIENCE

GGC-Trans: Gaze-Guided Cross-View Transformer for Calibration-Free Depth Generation

Dec 2025 – Present

Developed GGC-Trans, a Transformer-based architecture bridging egocentric semantic features with exocentric spatial coordinates.

Leveraged cross-attention mechanisms to eliminate the need for explicit camera calibration in ego-to-exo view synthesis.

Integrated gaze heatmaps as attentional priors in the Key-Value stream to localize user-attended objects in cluttered scenes.

Implemented a hybrid loss function combining L1 regression and gradient-based edge consistency to optimize depth sharpness.

Validated structural consistency using RealSense RGB-D ground truth through a custom-built depth-aware supervision pipeline.

BEAR: Benchmarking and Enhancing Multimodal LLMs for Atomic Embodied Capabilities

Jun 2025 – Sep 2025

Contributed to the development of a large-scale benchmark for skill-level evaluation of MLLMs in perception and reasoning tasks.

Led the implementation of baseline models and performed diagnostic analysis to identify spatiotemporal reasoning bottlenecks.

Co-developed BEAR-Agent, integrating visual-spatial tools to boost performance in both simulation and real-robot settings.

GazeCastGrasp / Eye-Gaze Interaction System for Assistive Robotic Manipulation

Feb 2025 – Sep 2025

Designed an end-to-end gaze-driven pipeline mapping user attention to robotic grasp targets in mixed reality environments.

Integrated eye-tracking with ROS control to enable intuitive, hands-free object selection for users with limited mobility.

Evaluated interaction efficiency through controlled user studies comparing gaze interaction against traditional input methods.

PUBLICATIONS

RaycastGrasp: Eye-Gaze Interaction with Wearable Devices for Robotic Manipulation, ICIR 2025

Oct 2025

Awarded **Best Paper at ICIR 2025**. Developed an MR interaction system for intuitive hands-free robotic control.

LSTM Neural Network-Based Forecasting of Civil Aviation Passenger Traffic, Exploration Science

Dec 2023

Designed a deep learning pipeline to capture long-term temporal patterns in large-scale transportation data.